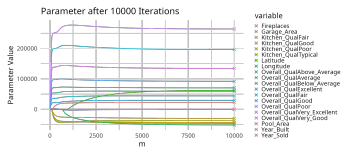
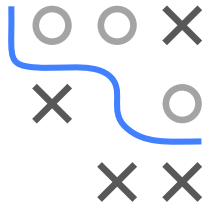


Introduction to Machine Learning

Boosting

Gradient Boosting: CWB and GLMs



Learning goals

- Understand relationship of CWB and GLM

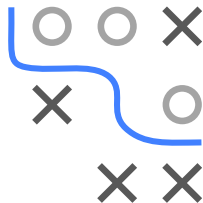
RELATION TO GLM

In the simplest case we use linear models (without intercept) on single features as base learners:

$$b_j(x_j, \theta) = \theta x_j \quad \text{for } j = 1, 2, \dots, p \quad \text{and with } b_j \in \mathcal{B}_j = \{\theta x_j \mid \theta \in \mathbb{R}\}.$$

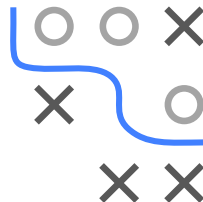
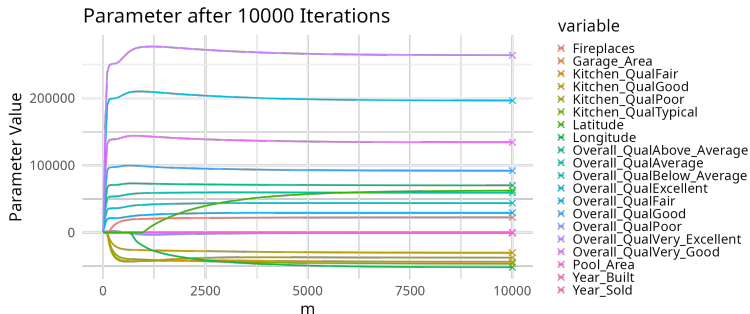
This definition will result in an ordinary **linear regression** model.

- In the limit, boosting algorithm will converge to the maximum likelihood solution.
- By specifying loss as NLL of exponential family distribution with an appropriate link function, CWB is equivalent to (regularized) **GLM**.



EXAMPLE: CWB PARAMETER CONVERGENCE

The following figure shows the parameter values for $m \leq 10000$ iterations as well as the estimates from a linear model as crosses (GLM with normally distributed errors):



Throughout the fitting of CWB, the parameters estimated converge to the GLM solution. The used data set is Ames Housing.